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From One-Time Standardization to Automated ETL: A Metadata-Driven Approach in SAS®

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ABSTRACT

After a large administrative dataset has been standardized, an important question remains: how can that standardized structure be maintained as new data arrive each month? This becomes particularly challenging when monthly vendor files contain millions of records but are accompanied by limited structural metadata. Without a consistent method for evaluating incoming files, structural differences can require repeated manual review and may introduce unintended changes into the master dataset.

This paper outlines a metadata-driven approach in SAS for managing recurring, large-scale administrative datasets. The framework establishes an internal metadata standard from the initial master dataset and uses it as the reference for subsequent monthly delta files. The metadata standard includes variable names, data types, and approved character lengths, with a controlled 10% buffer applied to character variables to accommodate reasonable growth while preserving structural consistency.

Delta files are ingested into SAS using vendor-provided maximum character widths to prevent data truncation. Each monthly file is then profiled to determine observed character lengths, and its structure is evaluated against the approved metadata standard. Files that conform to the standard proceed through the ETL process, while structural discrepancies are isolated for review before integration into the master dataset.

This approach provides a repeatable method for managing recurring data deliveries, identifies structural discrepancies early, and maintains a consistent master dataset. It also supports more efficient and scalable processing by replacing repeated file-by-file structural decisions with a standardized validation process.

INTRODUCTION

Standardizing a large administrative dataset is only the first step in creating a reliable analytical data environment. Once a standardized master dataset has been established, recurring deliveries must be processed in a way that preserves that structure over time. This is especially important when data supports analytical reporting, dashboards, and other downstream uses that depend on consistent variable definitions.

Metadata is essential for scaling, governing, and maintaining an effective ETL framework. Often described as “data about data,” metadata provides information about a dataset’s structure and characteristics and supports consistent processing and governance. Without sufficient metadata, ETL processes can become difficult to understand, maintain, and troubleshoot.

In a recent project involving the extraction of raw text files into analytical SAS datasets, insufficient metadata presented a significant challenge. Files were initially imported using vendor-provided maximum character widths to protect against data loss. Because the files contained millions of records, using these broad character widths increased processing demands and contributed to slower runtimes. This prompted the development of a more structured data governance approach to support an efficient and repeatable ETL process.

This was not a one-time ETL process; administrative data files were received on a recurring monthly basis. These deliveries contained millions of records and approximately 100 variables per file. Although the data represented the same underlying concepts from one delivery to the next, the technical structure of the files could not be assumed to remain unchanged. After the initial master dataset was created and standardized, the next challenge was determining how to maintain that standardized structure consistently as monthly delta files were delivered.

Limited information about the underlying data structures further complicated this process. In the environment described in this paper, source data originated from a SQL Server environment and were delivered as text files. The vendor provided a source-to-target mapping document, but character variables in the source system were broadly defined using generic widths such as VARCHAR(255) or VARCHAR(4000). These definitions provided a safe ingestion point but did not necessarily represent the most appropriate production metadata for the SAS analytical environment.

The result is a recurring ETL challenge. Every new delivery must be evaluated against the structure of the existing master dataset. Without a formal standard, differences may be addressed manually, resulting in inconsistent decisions, repeated quality control work, or unintended changes to the master dataset.

The objective of this paper is to describe a metadata-driven approach for managing this recurring process in SAS. Rather than treating metadata solely as documentation, the approach uses metadata as an executable control that determines whether an incoming dataset conforms to an established standard.

STRUCTURAL PROFILING AND METADATA STANDARDIZATION

Administrative data are received monthly in .txt format. Each monthly delivery consists of approximately 17 files, with each file containing more than one million records and about 100 variables. The vendor-provided source-to-target mapping supplies information needed to interpret each delivery, but it does not function as a complete field-level data dictionary for the SAS environment. Because documentation is limited, the focus of this work is the development of internal production metadata that defines how fields should be represented consistently within SAS.

Figure 1 provides a simplified representation of the data environment and ETL flow.

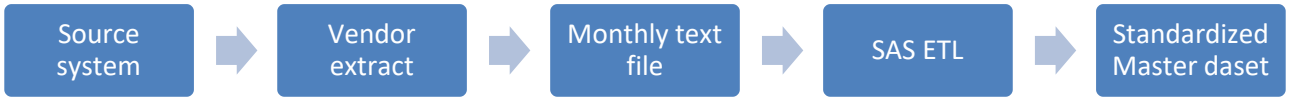


Figure 1. Simplified data environment and ETL flow.

CHALLENGE: MAINTAINING STANDARDIZED STRUCTURE

The initial standardization of a large dataset solves only one part of the ETL problem. Once a historical dataset has been standardized, subsequent delta files must continue to conform to that structure.

Table 1 illustrates how observed character lengths can differ between the master dataset and an incoming delta file.

Master Dataset Variable	Length	Delta Dataset Variable	Length
MEMBER_ID	20	MEMBER_ID	20
PROVIDER_NAME	41	PROVIDER_NAME	45
SVC_DESCRIPTION	70	SVC_DESCRIPTION	56

Table 1. Comparison of Character Lengths in the Master and Delta Datasets

Incoming delta files should not redefine the master structure each month simply because their observed character lengths are shorter or longer. The ETL process should not silently change the master dataset when an incoming delivery contains a structural discrepancy. Instead, the discrepancy should be identified and reviewed. This controlled process is a key component of the metadata framework and supports data governance by maintaining the structural integrity of the dataset over time.

ESTABLISHING THE METADATA STANDARD

The metadata standard was established from the standardized master dataset, with the goal of ensuring that future delta files could be integrated without allowing each delivery to redefine the master structure. Instead of allowing monthly deltas to control character lengths, the approved metadata serves as an internal governance standard. A 10% buffer was added to approved character lengths as a project-specific rule to accommodate reasonable future growth while retaining controlled structural boundaries. The monthly ETL process consists of the following steps:

1. Import .txt delta files
2. Profile and resize the delta
3. Compare the delta with the approved metadata standard
4. Combine the delta with the master dataset

STEP 1: IMPORT THE DELTA FILES

The delta is imported using the vendor's maximum character widths, which protects against truncation on ingest:

```
options compress=yes;
%LET LIB = S:\<Project>\Claims Data\Master SAS files\<YYYYMMDD> Monthly Update;
libname NEW "&LIB";

%MACRO COND(R1,R2,R3);
data NEW.&R1;
  infile "&R2" delimiter='||' MISSOVER DSD firstobs=2;
  informat SNPSHT_YR_AND_MTH_NBR BEST32.;
  informat MEMBER_ID $255.;
  informat RCRD_STRT_DT ANYDTDTE10.;
  informat RCRD_END_DT ANYDTDTE10.;
  informat RCRD_DLT_IND $255.;
  informat DATA_SRC_ID $255.;
  /* remaining vendor-max-width informats */
```

STEP 2: RESIZE THE DELTA TO ITS ACTUAL LENGTH

Each character variable is scanned for its true maximum length in the current delta, and a resized copy of the delta is built from that observed length instead of the vendor's maximum width:

```
%let data=nov.enroll;

data nov.nov_resize(keep=_name_ _length_ _format_ _informat_);
  set &data end=_eof;
  array _c[*] _character_;
  array _s[104] _temporary_;
  do _i_ = 1 to dim(_c);
    _s[_i_] = max(_s[_i_], length(_c[_i_]));  /* store max length observed */
  end;
  if _eof then do _i_ = 1 to dim(_c);
    length _name_ $32 _format_ _informat_ $10.;
    _name_ = vname(_c[_i_]);
    _length_ = _s[_i_];
    _format_   = strip(put(_length_,4.)) || ".";
    _informat_ = strip(put(_length_,4.)) || ".";
    output;
  end;
run;

/* Re-read the delta using the observed lengths */
data nov.enroll_resize_nov;
  %include FT23F001 / source2;
  set &data;
run;
```

STEP 3: COMPARE THE DELTA AGAINST THE METADATA STANDARD

The resized delta's structure is compared field by field against the approved standard. A full outer comparison identifies the ways in which a delta can diverge from the standard, including dropped variables, new or renamed variables, type mismatches, and length changes. Each discrepancy is assigned a category so that a reviewer can clearly identify what changed.

```
/* Pull column metadata for both datasets */
proc sql;
  create table work.master_cols as
  select upcase(name) as name, type, length
  from sashelp.vcolumn
  where libname='META' and memname='MASTER_STRUCTURE';

  create table work.delta_cols as
  select upcase(name) as name, type, length
```

```
  from sashelp.vcolumn
  where libname='NOV' and memname='DELTA_STRUCTURE';
quit;

proc sql;
  create table work.structure_diag as
  select
      coalesce(m.name, d.name)              as variable,
      m.type as master_type,   d.type as delta_type,
      m.length as master_length, d.length as delta_length,
      case
        when m.name is null                  then 'NEW_IN_DELTA'
        when d.name is null                  then 'MISSING_IN_DELTA'
        when m.type ne d.type                then 'TYPE_MISMATCH'
        when d.length > m.length              then 'LENGTH_EXCEEDS_STANDARD'
        else 'OK'
      end                                   as status
  from work.master_cols m
  full join work.delta_cols d
    on m.name = d.name;
quit;
```

A pass-or-fail flag is computed from the diagnostic table so that the pipeline, rather than a person reading a report, determines whether the delta can proceed:

```
proc sql noprint;
  select count(*) into :blocking_issues trimmed
  from work.structure_diag
  where status in ('NEW_IN_DELTA','MISSING_IN_DELTA','TYPE_MISMATCH',
                   'LENGTH_EXCEEDS_STANDARD');
quit;

%let structure_pass = %eval(&blocking_issues = 0);
```

A LENGTH_EXCEEDS_STANDARD result is treated as a blocking structural exception because it requires review before the delta can be integrated. The Managing Structural Exceptions section describes how these cases are resolved.

STEP 4: COMBINE THE DELTA WITH THE MASTER DATASET

Once a delta passes the Step 3 gate, it is combined with the master dataset.

```
%macro combine_delta;
  %if &structure_pass = 1 %then %do;
    proc sql;
      insert into meta.master_buffered (member_id, provider_name, svc_desc /*
...full column list... */)
      select member_id, provider_name, svc_desc
      from nov.delta_ready;
    quit;

    insert into meta.change_log
    values ("&sysdate9"d, "NOV_DELTA", 1, 0, "auto-approved: no blocking
issues");
  %end;
  %else %do;
    data meta.quarantine_&sysdate9._nov;
      set nov.delta_ready;
    run;
    %put WARNING: Delta blocked - &blocking_issues structural issues. Routed to
quarantine.;
  %end;
```

```
%mend;
%combine_delta;
```

METADATA AS AN ACTIVE ETL CONTROL

The central idea of the framework is to use the approved metadata as an automated control point within the ETL workflow rather than as a reference document consulted after the fact. Table 2 summarizes how an incoming file moves through that control point.

Table 2. ETL Control Flow for a Monthly Delta

Stage	Outcome if Structure Matches	Outcome if Structure Differs
Compare to approved standard	Conforms to standard	Held for review
Combine with master	Appended	Investigate and, if approved, version the standard

Every delta is imported with safe widths and then evaluated against the internal metadata standard before it is allowed to affect the master dataset. A match moves directly through the workflow; a mismatch is investigated, and the metadata standard changes only when that investigation results in an approved update.

VALIDATING THE MONTHLY DATA DELIVERY

Each monthly delta goes through four structural checks before it can be appended to the master file:

1. New variables: a variable appears in the incoming delivery that does not exist in the approved metadata.

2. Missing variables: a variable in the approved standard is absent from the incoming delivery, which may indicate a source-system change, an incomplete delivery, or another issue worth investigating.

3. Data type mismatch: the same variable is delivered with a different data type.

4. Length overage: a character variable exceeds its approved length. This is treated as a structural exception and routed for review rather than used to automatically expand the master dataset.

MANAGING STRUCTURAL EXCEPTIONS

Fields that do not conform to the approved metadata standard are held for review. The flagged variables are investigated to determine the cause of the discrepancy. The review then results in one of two outcomes:

- Approved expansion: If the overage reflects a legitimate, sustained change, such as provider names that have grown longer over several deliveries, the standard is updated. The change is versioned with an effective date, an approver, and a reason. The quarantined delta is then re-evaluated against the updated standard and appended normally.
- Rejected as an anomaly: If the overage appears to be a one-time data-entry error or vendor issue, the standard remains unchanged. The affected record is held out of the append and logged as an exception rather than being used to justify a permanent structural change.

The 10% buffer accommodates ordinary variation within the approved boundary. Anything beyond that boundary requires a deliberate review and approval decision, preventing a single unusual delivery from permanently reshaping the master structure.

CONFORMING THE INCOMING DATASET

Once a delta passes validation, the ETL process programmatically aligns its variables with the approved standard. For a dataset with hundreds of variables, writing that alignment logic manually for every field would defeat the purpose of a metadata-driven design. Instead, the approved metadata table is used to generate the required SAS statements. From that table, the process determines which variables exist, their data types, their approved lengths, and which variables require standardization or review. The same logic can then be reused across monthly deliveries.

A discrepancy is never allowed to change the approved metadata on its own. A length overage follows the same review-and-approval path described in the Managing Structural Exceptions section before the standard itself changes.

RESULTS

The framework's primary outcome has been a consistent dataset structure across recurring monthly deliveries. Incoming files are evaluated against the approved metadata standard before integration, and structural differences are surfaced as exceptions rather than propagated automatically into the master dataset. This keeps variable names, data types, and approved character lengths stable from one update to the next. That stability also reduces the burden on analysts, who can rely on a consistent dataset definition instead of adapting their work to structural differences introduced by an individual monthly delivery. In effect, metadata has moved from a descriptive artifact to an operational mechanism for maintaining structural consistency throughout the ETL lifecycle.

CONCLUSION

Recurring administrative data deliveries require more than a one-time effort to standardize a dataset. Once a master dataset is established, maintaining its structure across future deliveries becomes an ongoing ETL and data governance challenge in its own right.

This paper presented a metadata-driven approach in SAS that addresses this challenge by establishing an approved internal metadata standard and using it to validate every subsequent monthly delivery. Its key contribution is treating metadata as an operational ETL control rather than solely as documentation. Instead of requiring a manual structural decision for every incoming file, the process provides a reusable framework for validation, standardization, exception handling, and controlled, versioned changes to the standard itself.

CONTACT INFORMATION

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